METRIC

MIL-STD-1870A(AT)
10 December 1991
SUPERSEDING
(See 6.3)

MILITARY STANDARD

GRAIN SIZE, CAST TURBINE BLADES, ACCEPTANCE CRITERIA FOR (METRIC)

AMSC N/A FSC 2835

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FOREWORD

1. This Military Standard is approved for use by U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

3. The intent of this standard is to define grain size acceptance criteria for cast turbine blades.

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1. SCOPE

1.1 Scope. The procedures covered by this standard are to be used to insure cast turbine blades meet prescribed grain size acceptance criteria.

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3. DEFINITIONS

3.1 Definitions. For the purposes of this standard, the following definitions shall apply.

3.1.1 Grain size. The measurements made to determine grain size are made on surface of casting. Dimensional limitations specified herein are defining grain structure as a surface phenomena. Internal grain size is excluded from requirements specified in 4.1. However, it is expected that a grain occurring at the juxtaposition of two or more planes will be measured separately in each plane where it appears.

3.1.2 Airfoil shape. An airfoil shape is a casting or portion of a casting which extracts work or changes direction of the hot gas stream in the turbine section of a gas turbine engine.

3.1.3 Other shapes. Other shapes are any casting or portion of a casting which cannot be defined as an airfoil shape (e.g., root area or tip shroud of a cast blade).

3.1.4 Casting lot. Unless otherwise specified in applicable process specifications, a casting lot is a group of castings with the same part number and revision letter poured from one master heat of metal using the same casting process parameters and the same type of preheat and casting furnace.

3.1.5 Columnar grains. A columnar grain shall be any grain which has its maximum dimension at least four times the length of its minimum dimension. Any columnar grain 3.180 millimeter (mm) (0.125 inch) maximum dimension or less shall not be considered ratable.

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4. GENERAL REQUIREMENTS

4.1 Acceptance criteria. The grain size shall not exceed the following requirements:

4.1.1 Airfoil shapes. The grain size shall be generally equiaxed and uniform throughout the airfoil (see 3.1.2) and shall not exceed 3.180 mm (0.125 inch).

4.1.2 Other shapes. The grain size in other cast shapes (see 3.1.3) shall be generally equiaxed and uniform throughout the section and shall not exceed the measurements shown in table I.

TABLE I. Casting thickness and grain size.

Casting thickness		Grain size	
mm	inch	mm	inch
0 - 3.180	0 - 0.125	0 - 3.180	0 - 0.125
3.180 - 12.700	0.125 - 0.500	4.750	0.187
over 12.700	0.500	6.350	0.250

4.1.3 Columnar grains. Columnar grains (see 3.1.5) shall not be acceptable except for those areas designated in figure 1.

4.1.4 Transition zones. Unless otherwise specified in the engineering drawing, sharp areas of transition in grain size shall not be cause for rejection.

4.2 Responsibility for inspection. The contractor is responsible to conduct and complete a sufficient inspection to ensure:

a. Compliance with all of the requirements in this standard.
b. That adequate traceability exists to verify that the requirements of this standard have been meet.
c. That the integrity of castings is equivalent in all other respects to sound acceptable castings.

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5. DETAIL REQUIREMENTS

5.1 Production in process control. The contractor shall ensure that the supplier identifies the part number of the casting item to the following (see 6.3):

a. Engineering drawing and revision.
b. Quantity of items in the lot.
c. Lot number.
d. Applicable issue and revision of this standard.

5.2 Frequency of inspection. A minimum of the first 500 blades per casting lot from the first three production casting lots (see 3.1.4) shall be grain size inspected in accordance with the applicable engineering drawings and specifications requirements. After three successive production casting lots have been accepted by this method, the succeeding casting lots may be grain size inspected to a sampling plan approved by the acquisition activity. The acceptance number in all cases shall be zero.

5.2.1 Reinspection. If a production lot of blades is not 100 percent (%) acceptable, the entire lot of blades represented by the initial 500 blades shall be grain size inspected. When any sample lot of blades exceeds the approved rejection rate, the balance of the lot represented by the sample lot shall be grain size inspected.

5.3 Test method. Each casting shall be 100% grain size etched to verify conformance to 4.1. Etchant used shall conform to the applicable engineering drawings and specifications requirements.

NOTE: After completion of grain size inspection, etched castings may be cleaned by light abrasive blasting.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to be used to ensure cast turbine blades meet prescribed grain size acceptance criteria.

6.2 Data requirements. Although this standard does not include any requirements for data, it does not exclude contractors from obtaining data from its suppliers as required to ensure compliance to this standard and to verify conformance to the applicable engineering drawings.

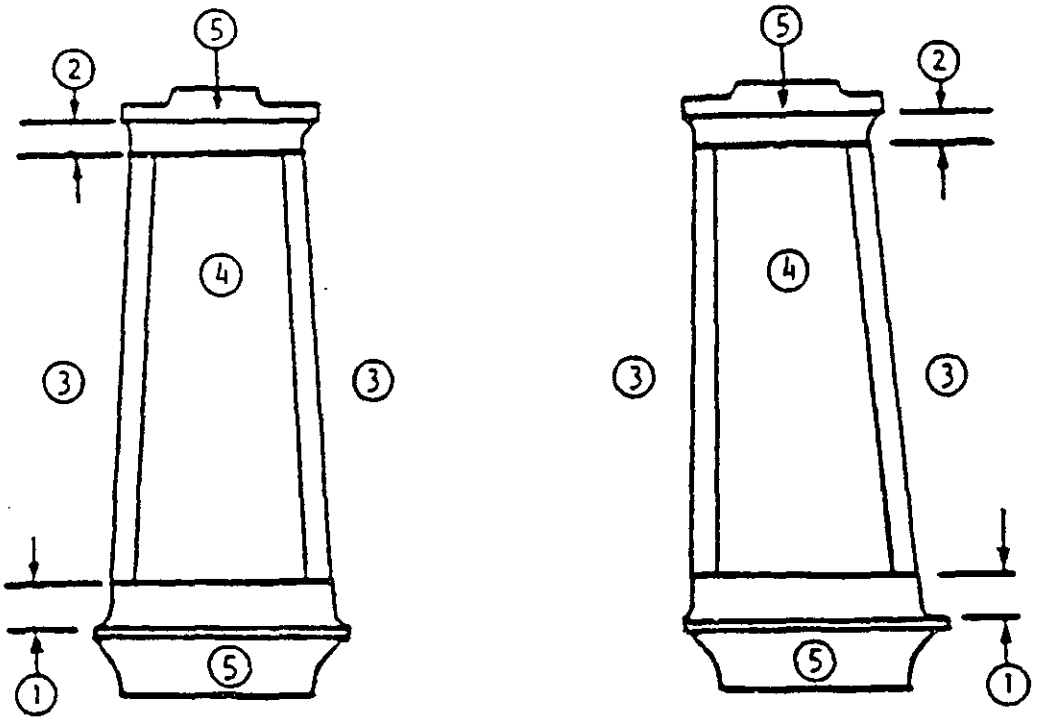
6.3 Supersession data. This standard supersedes MIL-STD-1870(AT), dated 13 May 1983 and Textron Lycoming specification P7075D, dated 3 February 1983.

6.4 Subject term (key word) listing.

Abrasive blast
Airfoil shape
Casting lot
Columnar grains
Equiaxed
Grain size
Juxtaposition
Shroud
Transition zones
Turbine blade

6.5 Changes from previous issue. Vertical lines or asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTES:

1. Casting Areas:

   (1) 15% of blade length at the trailing edge.

   (2) 20% of blade length at the trailing edge.

   (3) 10% of mean chordal width.

   (4) Airfoil surface as shown.

   (5) See 4.1.2.

2. Grain Size Allowable:

   Areas (1) and (2) : Columnar grains up to 12.700 mm (0.500 inch) maximum dimension not exceeding an angle of 45 degrees to the stacking line shall be permitted. Not more than 1/5 of this area shall contain columnar grains.

   Areas (3) and (5) : No ratable columnar grains shall be permitted.

   Area (4) : Columnar grains up to 9.500 mm (0.370 inch) maximum dimension shall be permitted.

FIGURE 1. Casting areas for grain size allowance.

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